GREATEST COMMON DIVISORS OF THE NUMBERS $2^n - 1$

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Abstract. We give an elementary proof that $\gcd(2^m - 1, 2^n - 1) = 2^{\gcd(m,n)} - 1$ for all natural numbers m, n, by running the Euclidean algorithm on the exponents. We derive the usual consequences for coprimality and for prime values, and discuss the prime divisors of $2^p - 1$.

1. Introduction

The sequence $2^n - 1$ is a *strong divisibility sequence*: the gcd of two of its terms is again a term, with index the gcd of the indices. This property was studied systematically by Lucas [1]. We present a short self-contained proof (Theorem 2.6) based on a single addition formula (Lemma 2.2), and collect some consequences.

2. The gcd identity

Definition 2.1. For $n \in \mathbb{N}$ put $a_n = 2^n - 1$. Thus $a_0 = 0$, $a_1 = 1$, $a_2 = 3$, $a_3 = 7$, $a_4 = 15$. We use the convention $\gcd(m, 0) = m$ for $m \in \mathbb{N}$.

Lemma 2.2. *For all $m, n \in \mathbb{N}$ we have $a_{m+n} = 2^n a_m + a_n$.*

Proof. $2^n(2^m - 1) + (2^n - 1) = 2^{m+n} - 1$. □

Lemma 2.3. *For every $n \in \mathbb{N}$ the number a_n is odd.*

Proof. 2^n is even, so $2^n - 1$ is odd. □

Lemma 2.4. *Let m, n be integers with $1 \le m \le n$. If $m \mid n$, then $a_m \mid a_n$.*

Proof. Write $n = mq$. Then $a_n = (2^m)^q - 1 = (2^m - 1)\big(1 + 2^m + 2^{2m} + \cdots + 2^{(q-1)m}\big)$. □

Lemma 2.5. *Let $m \ge n \ge 0$. Then $\gcd(a_m, a_n) = \gcd(a_{m-n}, a_n)$.*

Proof. By Lemma 2.2 applied to the pair $(m-n, n)$ we have $a_m = 2^m a_{m-n} + a_n$, hence $\gcd(a_m, a_n) = \gcd(2^m a_{m-n}, a_n)$. Since a_n is odd by Lemma 2.3, it is coprime to 2^m, and therefore $\gcd(2^m a_{m-n}, a_n) = \gcd(a_{m-n}, a_n)$. □

Theorem 2.6 (Main theorem)**.** *For all $m, n \in \mathbb{N}$,*

$$\gcd(2^m - 1,\, 2^n - 1) = 2^{\gcd(m,n)} - 1.$$

Proof. We argue by strong induction on $m+n$. If $n = 0$, then $\gcd(a_m, a_0) = \gcd(a_m, 0) = a_m = a_{\gcd(m,0)}$, and the case $m = 0$ is symmetric. Otherwise we may assume $m \geq n \geq 1$ by symmetry. Then by Lemma 2.5 and the induction hypothesis applied to $(m-n, n)$,

$$\gcd(a_m, a_n) = \gcd(a_{m-n}, a_n) = a_{\gcd(m-n,n)} = a_{\gcd(m,n)}. \qquad \square$$

Corollary 2.7. *For $m, n \in \mathbb{N}$, the numbers $2^m - 1$ and $2^n - 1$ are coprime if and only if $\gcd(m, n) = 1$.*

Proof. By Theorem 2.6 the gcd is a_g with $g = \gcd(m, n)$, and $a_g = 1$ if and only if $g = 1$. $\square$

Proposition 2.8. *Let $m, n \geq 1$ be integers such that $m \nmid n$ but $a_m \mid a_n$. Then $n > m$.*

Proof. Since $a_n \geq 1$ and $a_m \mid a_n$ we get $a_m \leq a_n$, so $m \leq n$; and $m \neq n$ because $m \nmid n$. $\square$

3. Prime values and prime divisors

Proposition 3.1. *If a_n is prime, then n is prime.*

Proof. As $a_0 = 0$ and $a_1 = 1$, we have $n \geq 2$. If $n = rs$ with $1 < r < n$, then $a_r \mid a_n$ by Lemma 2.4 and $1 < a_r < a_n$, so a_n is composite. $\square$

Proposition 3.2. *Let p be an odd prime and q a prime divisor of a_p. Then $q \equiv 1 \pmod{4p}$.*

Proof. Since a_p is odd, q is odd. The multiplicative order of 2 modulo q divides p and is not 1, hence equals p. By Fermat's little theorem it also divides $q - 1$, so $p \mid q - 1$; as $q - 1$ is even and p is odd, $2p \mid q - 1$. $\square$

Remark. By Zsigmondy's theorem [2], for every $n \geq 2$ the number a_n has a prime divisor that divides none of $a_1, \ldots, a_{n-1}$. In view of Theorem 2.6, such a *primitive* prime divisor of a_n divides no a_k with k not a multiple of n.

Conjecture 3.3. *There are infinitely many n for which a_n is prime.*

References

[1] É. Lucas, *Théorie des fonctions numériques simplement périodiques*, Amer. J. Math. **1** (1878), 184–240, 289–321.

[2] K. Zsigmondy, *Zur Theorie der Potenzreste*, Monatsh. Math. Phys. **3** (1892), 265–284.

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